

SYDNEY PENRITH, Australia

WMO: 947630

Lat: 33.7194S

Lon: 150.6783E

Elev: 81

StdP: 14.65

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.4	39.3	27.9	22.0	62.6	30.4	24.6	59.9	23.9	58.7	20.2	60.5	1.9	190	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	98.3	69.8	93.5	69.5	89.5	68.7	75.2	85.9	73.7	83.1	72.5	81.5	10.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.0	122.9	77.2	71.5	116.8	75.6	70.1	111.2	74.9	38.6	86.2	37.3	82.9	36.2	81.4	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.3	17.0	14.4	DB	33.2	106.6	1.7	4.7	32.0	110.0	31.0	112.7	30.1	115.3	28.8	118.7
			WB	32.7	78.2	1.6	2.0	31.5	79.6	30.6	80.7	29.7	81.8	28.6	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	75.9	74.3	71.4	65.6	58.9	54.4	52.9	55.0	60.8	65.6	69.4	73.6	(6)
	DBStd	9.25	5.68	5.37	4.43	4.23	3.86	3.49	3.41	3.67	4.72	5.53	6.12	5.66	(7)
	HDD50	18	0	0	0	0	0	5	10	3	0	0	0	0	(8)
	HDD65	1472	0	1	3	41	190	319	376	310	144	61	23	4	(9)
	CDD50	5407	803	682	663	469	278	136	99	158	324	483	582	730	(10)
	CDD65	1387	338	263	201	60	3	0	0	1	17	79	156	269	(11)
	CDH74	11189	2844	1917	1190	351	26	0	4	15	293	860	1401	2288	(12)
	CDH80	4689	1362	797	374	76	1	0	0	1	79	333	613	1053	(13)
(14) Wind	WSAvg	5.6	6.4	6.2	5.5	4.9	4.3	4.7	4.7	5.5	5.9	6.1	6.5	6.2	(14)
(15) Precipitation	PrecAvg	34.8	4.0	4.3	3.9	3.0	2.3	2.4	1.5	2.0	1.6	2.6	3.7	3.4	(15)
	PrecMax	55.4	12.1	17.6	12.0	12.4	8.5	11.1	6.8	15.7	7.6	6.7	21.3	10.6	(16)
	PrecMin	17.6	0.2	0.4	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	(17)
	PrecStd	10.3	3.0	4.0	2.8	3.0	2.3	2.7	1.7	3.1	1.4	2.0	3.9	2.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.1	101.6	93.6	88.0	77.5	69.8	72.5	75.9	88.0	94.1	100.1	101.9	(19)
		MCWB	72.4	72.3	70.4	65.5	60.3	57.6	56.3	56.7	61.7	63.5	65.9	69.5	(20)
	2%	DB	98.2	94.2	87.9	81.4	73.4	66.4	67.2	71.6	81.7	88.3	92.6	95.4	(21)
		MCWB	71.5	72.1	70.2	64.6	60.3	55.6	53.6	55.7	59.7	63.2	67.0	68.9	(22)
	5%	DB	92.4	88.6	83.9	77.9	70.6	64.3	64.2	68.2	76.8	83.2	86.6	90.2	(23)
		MCWB	70.8	71.3	68.8	64.2	58.8	54.7	52.8	54.1	59.0	62.5	65.7	67.9	(24)
	10%	DB	87.5	84.4	80.4	74.7	68.0	62.2	61.8	65.2	72.7	78.1	81.9	85.7	(25)
		MCWB	70.1	69.4	67.7	63.2	57.6	53.9	51.8	52.7	57.6	61.3	64.9	67.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.1	77.9	75.5	70.2	65.6	61.4	60.8	60.9	64.9	69.3	73.4	75.3	(27)
		MCDB	89.0	90.4	82.4	77.7	70.2	64.8	64.8	68.7	75.1	79.5	83.7	86.7	(28)
	2%	WB	74.9	75.4	73.1	68.1	63.2	59.1	57.0	58.2	62.9	66.9	70.9	73.0	(29)
		MCDB	87.6	84.5	80.2	74.9	68.4	62.2	61.8	66.3	73.6	77.0	81.1	84.1	(30)
	5%	WB	73.5	73.7	71.5	66.6	61.2	57.7	54.9	56.3	61.2	65.3	69.2	71.5	(31)
		MCDB	83.6	81.6	78.4	72.9	66.9	60.9	59.9	64.1	71.1	75.8	78.7	81.9	(32)
	10%	WB	72.3	72.1	70.0	65.2	59.7	56.2	53.6	54.6	59.4	63.7	67.5	70.1	(33)
		MCDB	82.0	80.2	76.7	71.3	65.3	59.7	59.1	62.0	69.3	73.5	76.5	80.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	17.8	17.2	18.7	19.7	17.8	20.3	22.2	23.2	22.3	20.4	20.9	(35)
		MCDBR	31.2	26.8	23.9	24.1	23.2	20.9	24.6	27.7	31.4	32.6	31.3	30.3	(36)
	5% WB	MCWBR	8.3	8.0	8.4	9.4	11.0	11.4	12.7	12.9	11.9	10.9	9.3	8.7	(37)
		MCWBR	23.5	20.8	20.0	19.6	19.0	15.5	19.2	23.1	25.4	24.2	23.7	24.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.396	0.393	0.364	0.334	0.317	0.307	0.297	0.304	0.337	0.360	0.386	0.392	(40)
		taud	2.428	2.459	2.517	2.574	2.567	2.598	2.607	2.579	2.486	2.449	2.421	2.419	(41)
	Ebn at Noon		299	293	291	283	271	266	276	289	294	300	300	301	(42)
		Edh at Noon	39	36	32	27	24	22	23	26	32	36	39	40	(43)
(44) All-Sky Solar Radiation	RadAvg		2026	1725	1480	1210	970	767	898	1184	1571	1806	1902	2087	(44)
	RadStd		161	166	137	106	67	67	66	74	95	182	182	205	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (37 neighbors)	+0.79	N/A	N/A	+1.39	N/A	N/A	-4	-155	+333	+178	(47)

Nomenclature: See separate page